Track 2 report — iteration 4: one mechanism read from many independent sources, and a drug list tested against chance

Standing: ready, written 2026-09-25 from executed runs of iteration 4 only (steps 1–11). It does **not** use the results of any earlier iteration: the candidate list, the directions and the checks were all rebuilt from iteration-4 evidence. The one input taken from outside iteration 4 is the challenge's own answer: the oracle confirmed the causal pair (entry 083). Every number cites the methodology entry that measured it. Every claim was registered before its test. **Not uploaded.** Track 2 accepts one submission (due about 2026-10-20).

Written for a panel of clinicians, patient advocates and scientists. Each technical term appears once with its meaning in brackets; the full dictionary is served at `/learn` on the dashboard.

> The challenge's own words, kept in view throughout: *submissions are hypotheses for follow-up, not evidence that a medicine works.* Nothing below is a recommendation to treat.

1. How this report answers the panel's four questions

> **Rubric:** all four. This is the map; each row names the section that carries the work and the weakest point we know in it.

criterion	weight	where	the evidence, in one line	the weakest point we know
Scientific rigor	35 %	§3–§6	the mechanism is read from **five instruments that share no data** (variant-effect models, a structure predictor, a stability atlas, 76,699 mouse genotypes, a gene ranking on the clinical terms); four point the same way and one is inconclusive; the drug list comes from **four generators fixed before any join**, and their agreement is **tested against a permutation null** instead of assumed; 12 claims filed before their runs; **5 came back refuted or uninterpretable** and are reported as such (§12)	the four generators agree on **one** drug, and chance gives one (p 0.263, H094 refuted). The survivor, dasatinib, was known before the join. Two of the four generators are weak: the screen sweep finds nothing past correction, and the expression-signature generator failed its own positive control
Potential impact	25 %	§7	three things a treating team can act on **for this child**: a named pharmacogenomic test (the delivered callset answers 1.6 % of the positions it needs); vincristine flagged by **both** safety sources; an incidental malignant-hyperthermia susceptibility call. Plus the mouse model any preclinical test should use	no candidate has been tested in this child or in any BubR1-deficient model with an approved drug; "unflagged" means two sources are silent, not that a drug is safe
Innovation	25 %	§8	convergence of independent drug generators **tested against chance**; a patient-to-mouse-**genotype** phenotype match over a whole database; a per-child safety check that reads the child's **genome and clinical terms side by side**; a stability atlas **calibrated on published alleles before** the child's allele is placed	every part uses public tools; what we claim is the assembly and the discipline, "to our knowledge". We ran no systematic survey of prior art
Scalability	15 %	§9	every stage writes one table contract with a provenance manifest; the whole chain **ran on a second disease** (Bloom syndrome), $0 in 53 minutes, and proposed a different drug set (10 of 105 drugs shared)	our own claim that it would move **by configuration alone** came back **refuted** (H096): four stages kept the case in code and seven wrote the first case's text into the second's outputs. A 378-line driver fixed both, and names exactly what to fold into the tools next

2. The answer in one page

> **Rubric:** Impact and Rigor. What a reader who stops here should take away.

The mechanism (methods form Q14). Loss of function. The child carries two different broken copies of `BUB1B` (a **compound heterozygote**): one `stop_gained` allele (the protein is cut short) and one missense allele (one amino acid swapped). The missense is predicted to **destabilise** BubR1, the protein `BUB1B` makes. The result is too little stable BubR1, a weakened **spindle assembly checkpoint** (the cell's "all chromosomes attached?" check before division), and so cells with the wrong number of chromosomes (**aneuploidy**). Four of five independent instruments support this reading; the fifth, a structure prediction, is inconclusive (§3).

The drugs.

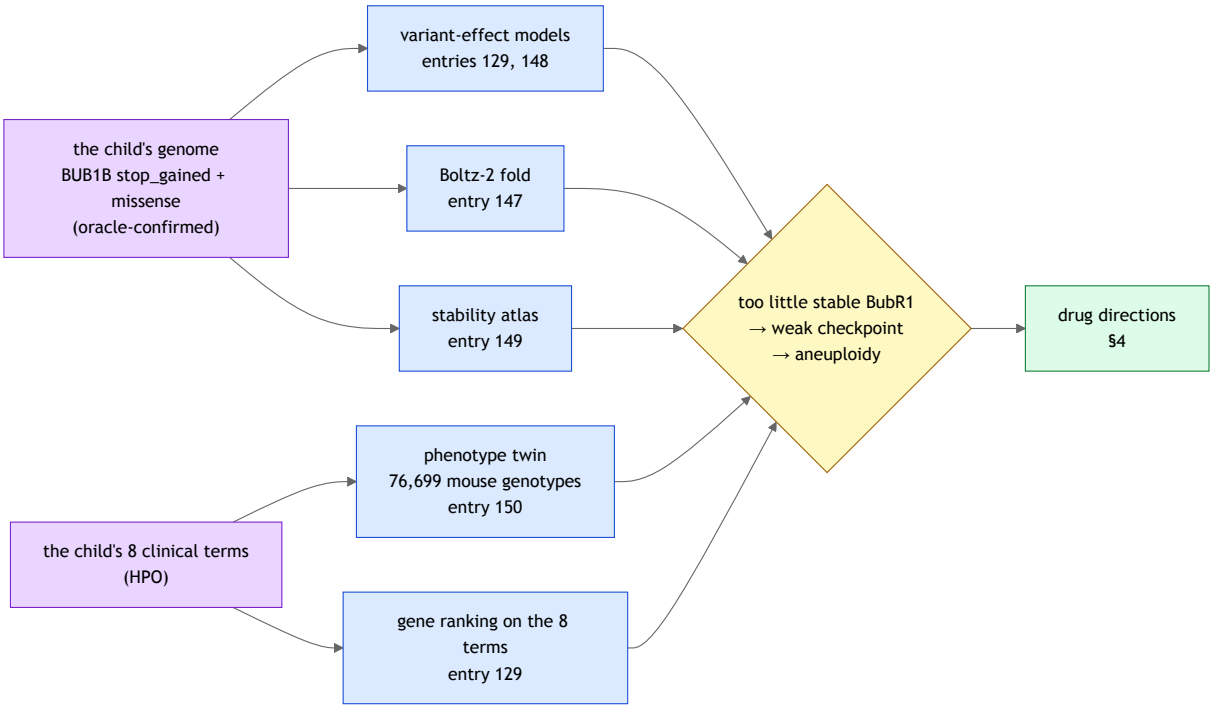
	drug	status	why	the caveat
the one survivor	**dasatinib**	approved (leukaemia)	named by two generators that share no data: the child's phenotype twin (a BubR1-deficient mouse whose ageing was delayed by clearing senescent cells) and a cancer-cell screen (99.4th percentile for killing aneuploid lines); passes every check	the two agree **at chance level** (p 0.263); it was known before the join; senolytic use is investigational; paediatric reports pair it with growth retardation
the single-source lead	**niacin** (vitamin B3)	approved	the twin's other rescue: raising NAD^+/SIRT2 raised BubR1 and extended life in the hypomorph mouse	one generator only; never tested with an approved drug in a BubR1 model
do not	**Mps1/TTK and Aurora kinase inhibitors**	none approved	they weaken the same checkpoint further; in 919 cancer lines 33 of 34 are *less* effective in aneuploid lines	none is in any candidate list; stated so no one adds one
flag	**vincristine**	approved, a standard rhabdomyosarcoma drug	the one drug **both** safety sources flag for this child: an observed CYP3A5 poor-metaboliser call, and its label names the child's own skeletal muscle atrophy	a monitoring question for the treating team, not a removal

The one action. Order a targeted pharmacogenomic panel (or re-call the reads) over SLCO1B1, ABCG2, CYP2C9, G6PD and UGT1A1. The delivered callset cannot answer any of them (§7.1).

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3. The mechanism (methods form Q14), from five instruments that share no data

> **Rubric:** Rigor (35 %). The mechanism is the basis of every drug argument, so it is read from as many independent instruments as iteration 4 could run.



> **Legend** purple = the raw inputs (the genome and the clinical terms) · blue = one independent instrument each · yellow = the conclusion they are weighed into · green = where it leads.

3.1 The genotype

fact	value	source
the causal pair	`BUB1B` `stop_gained` + missense, in trans (one on each parental copy)	oracle reply, full match at rank 1 (entry 083)
the pair in iteration 4's normalised records	`BUB1B` carries exactly one `missense_variant` and one `stop_gained`	entry 114

3.2 What the variant-effect models say (steps 1–8, "the model zoo")

A **variant-effect model** scores how damaging one DNA change is. Iteration 4 ran every credible open model it could license under one contract.

model	what it scores	the child's alleles	source
AlphaMissense, ESM1b, REVEL	missense only	the missense allele in their top **0.47 %, 2.52 %, 2.59 %** of everything scored	entry 129
GPN-MSA (a DNA-level model)	any single-letter change	the pair ranks **508th of 54,865** (0.93rd percentile); missense 1.38 %, stop-gained 4.09 %	entry 148
Evo 2 (a DNA language model, GPU)	any change	the stop-gained half at 5.61 %	entry 148
the child's 8 clinical terms alone	genes, not variants	`BUB1B` ranks **#14 of 5,267** genes	entry 129

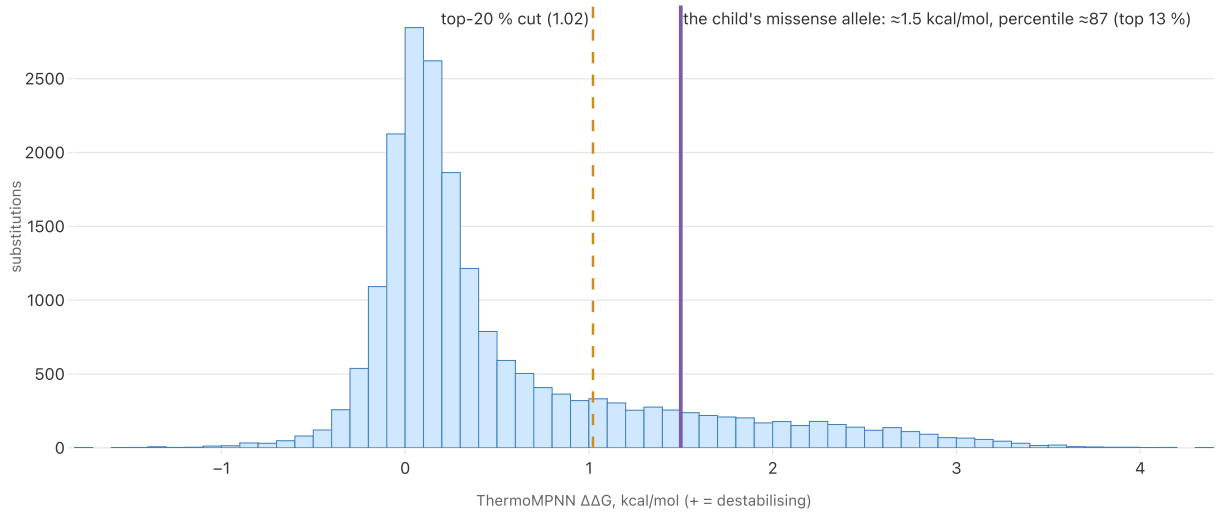
Fusing all models into one ranking does **not** find the pair (rank 20,639 of 52,012 with four models, 4,318 with nine; entries 130, 148): most models cannot score a nonsense allele, so they abstain. The per-allele reading

is still clear: both alleles are damaging.

3.3 Is the missense protein unstable or inactive? (structure, stability)

instrument	result	what it supports	source
Boltz-2 fold (a structure predictor), wild type vs mutant	a local confidence dip of ~3 pLDDT points (pLDDT: the predictor's own 0–100 confidence), the same size as a conservative swap at that residue	nothing either way; inconclusive (H081)	entry 147
stability atlas: ΔΔG (the change in folding energy; + = destabilising) for all **19,950** possible BubR1 substitutions (ThermoMPNN)	the child's missense at the **87th percentile**, buried, confidently folded (pLDDT ≥ 90), inside the pseudokinase domain	"predicted destabilising" (H084 supported, 0.55)	entry 149
calibration before placing the child's allele: can ΔΔG separate published abundance-lowering MVA alleles (5) from stable ones (3)?	AUROC **0.60** (95 % CI 0.13–1.00); no power at n = 8	the class is a **prediction**, not a result (H083 refuted, 0.15)	entry 149

ΔΔG of all 19,950 possible BubR1 substitutions



> **Legend** blue bars = every possible substitution (neutral machinery) · yellow dashed = the pre-registered top-20 % cut (a decision point) · purple = the child's allele (the raw input), value rounded to 0.1.

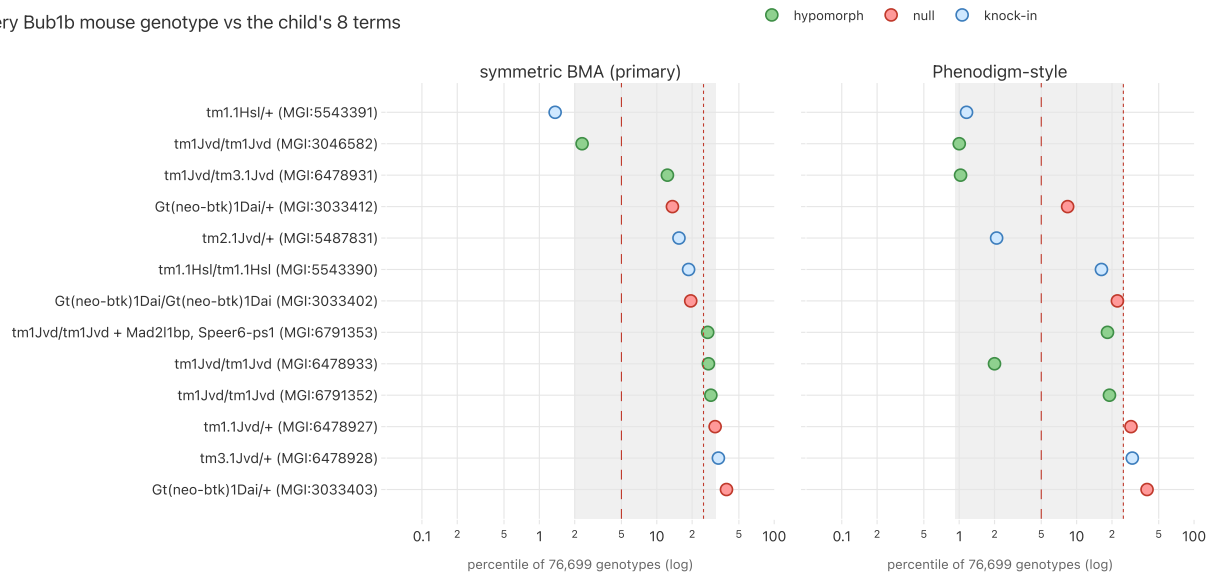
The `stop_gained` allele makes no full-length protein either way. So total stable BubR1 is low under both readings of the missense.

3.4 Which mouse is this child? (the clinical terms, independent of the genome)

The child's 8 clinical terms were scored against **all 76,699** mouse genotypes in MGI (the Mouse Genome Informatics database) through the mouse–human phenotype bridge (entry 150).

result	value
best `Bub1b` genotype among 76,699	top **1.0–1.4 %** (two scoring rules), ~10× better than random terms of equal specificity (p 0.026 and 0.060)
positive controls	Bloom syndrome → Blm mice passes; 6 of 7 calibration diseases in the top 5 %
which `Bub1b` mouse	the **hypomorph** (an allele that makes less working protein: BubR1 H/H, H/L1002P) on growth, muscle and perinatal terms; an acetylation-deficient knock-in on the tumour term
verdicts	H085 supported (0.70); H086 undecided (0.45): the two rules disagree on the single best mouse

Every Bub1b mouse genotype vs the child's 8 terms



> **Legend** green = hypomorph (H086's prediction) · red = null / knockout · blue = knock-in · grey band = context, not the subject: the central 90 % of the null · dashed red = 5 % (H085's prediction) · dotted red = 25 % (its refutation).

The mechanism, stated for the form. Loss of function, of the hypomorphic kind: too little stable BubR1 (a truncating allele plus a predicted-destabilising missense); pathway disrupted: the spindle assembly checkpoint (BubR1 is part of the mitotic checkpoint complex that holds anaphase until every chromosome is attached); downstream consequence: chromosome mis-segregation, mosaic aneuploidy, and in the mouse model early senescence, muscle wasting and growth failure, which match the child's own terms. No source suggests gain of function or a dominant-negative protein.

4. From mechanism to drug directions

> **Rubric:** Rigor. Each direction follows from the mechanism in §3, and each names the iteration-4 evidence that supports or weakens it.

id	direction	the logic	iteration-4 evidence	standing
D3	**raise stable BubR1**	a hypomorph makes some working protein; more of it helps	twin: SIRT2 overexpression / NAD^+ raised BubR1 and extended life in the H/H mouse; SIRT2 does nothing on a normal background (entry 150). Missense predicted destabilising (entry 149)	supported by the twin; no approved drug tested
D5	**clear senescent cells / damp mTOR**	low BubR1 drives early senescence; clearing $p16^+$ cells delayed it in the twin	twin: genetic clearance of $p16^+$ cells delayed sarcopenia and fat loss in H/H (entry 150); dasatinib is the approved senolytic analogue; the screen ranks dasatinib 99.4th percentile on aneuploid lines (entry 151)	two sources, one drug; pharmacological test in the twin never published
D2	**exploit aneuploidy stress**	aneuploid cells carry protein and energy stress	no Launched drug selective for aneuploid lines past correction (best q 0.287, entry 154); the one aneuploidy-ranked survivor, dasatinib, is argued under D5	not supported by iteration 4
D4	**the tumour context** (rhabdomyosarcoma)	the child's presenting tumour	no Launched drug selective for rhabdomyosarcoma lines past correction (best q 0.122; three lines carry most compounds, entry 154)	not supported; underpowered
A1	**do not: weaken the checkpoint further**	the child's checkpoint is already weak	33 of 34 Mps1/Aurora inhibitors are *less* effective in high-aneuploidy lines (median δ –0.139, entry 151), reproducing Cohen-Sharir 2021 (PMID 33505028)	measured; carried as check K1
A2	**caution: spindle poisons** (tubulin drugs)	they act through the checkpoint the child lacks	biology only in iteration 4; vincristine separately flagged by the safety check (§6.2)	a caution, not an exclusion

5. How the candidates were found: four generators that share no data

> **Rubric:** Rigor and Innovation. A candidate list is only as good as the independence of the sources behind it, so iteration 4 made that independence testable.

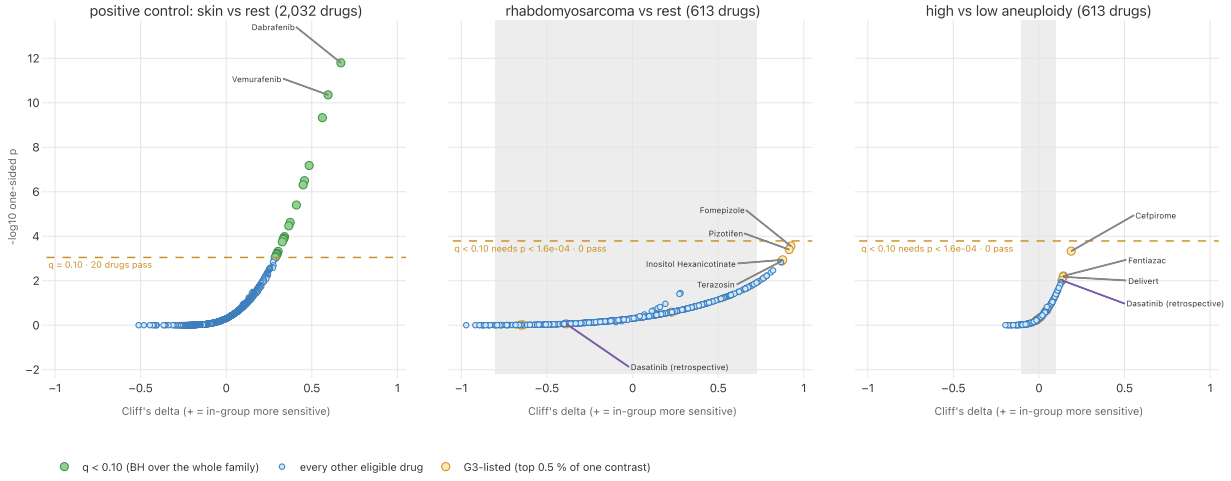
A **generator** proposes drugs by a rule fixed before any join. A **check** can only filter. The rules were committed before two of the four lists existed (commit `1b02e44`).

id	generator	the rule	listed / approved universe	source
G1	walk over PrimeKG (a biomedical knowledge graph) from the MVA disease node	top 50 labelled drugs by walk score	50 / 188	entry 142, 155
G2	the phenotype twin's published rescues	approved analogues of interventions that rescued the twin	2 / 1,498 (dasatinib, niacin)	entry 150, 155
G3	screen-wide PRISM sweep (a pooled cancer-cell drug screen, 613 approved drugs)	top 0.5 % on each contrast	8 / 613	entry 154
G4	LINCS signature reversal (drugs whose gene-expression effect is the opposite of losing `BUB1B`)	approved drugs in both engines' top 100	3 / 1,157	entry 153

Two generators are weak, and the report says so.

- **G3.** No drug passes multiple-testing correction in either contrast (H093 refuted, 0.10). The same code recovers the textbook melanoma drugs (dabrafenib at q 3.2×10^{-9}), so the pipeline works; the biology is not there at this power (entry 154).
- **G4.** Its registered positive control failed: the `BUB1B`-loss signature did not resemble the loss of the other checkpoint genes (best, BUB1, at 5.6 % against a 5 % cut). It reads as a generic cell-cycle arrest, so H092 is **uninterpretable** and G4's three drugs carry a caution (entry 153).

S1 — the same test finds BRAF inhibitors screen-wide, and nothing for the child's two contrasts



> **Legend** green = q < 0.10 · yellow = G3-listed · blue = every other eligible drug · grey band = central 95 % of δ over 200 seeded unrelated drugs · dashed yellow = the p needed for q < 0.10 (control: the largest passing p; real contrasts: 0.10 / 613, which no drug reaches) · purple label = dasatinib, marked only because entry 151 reported it.

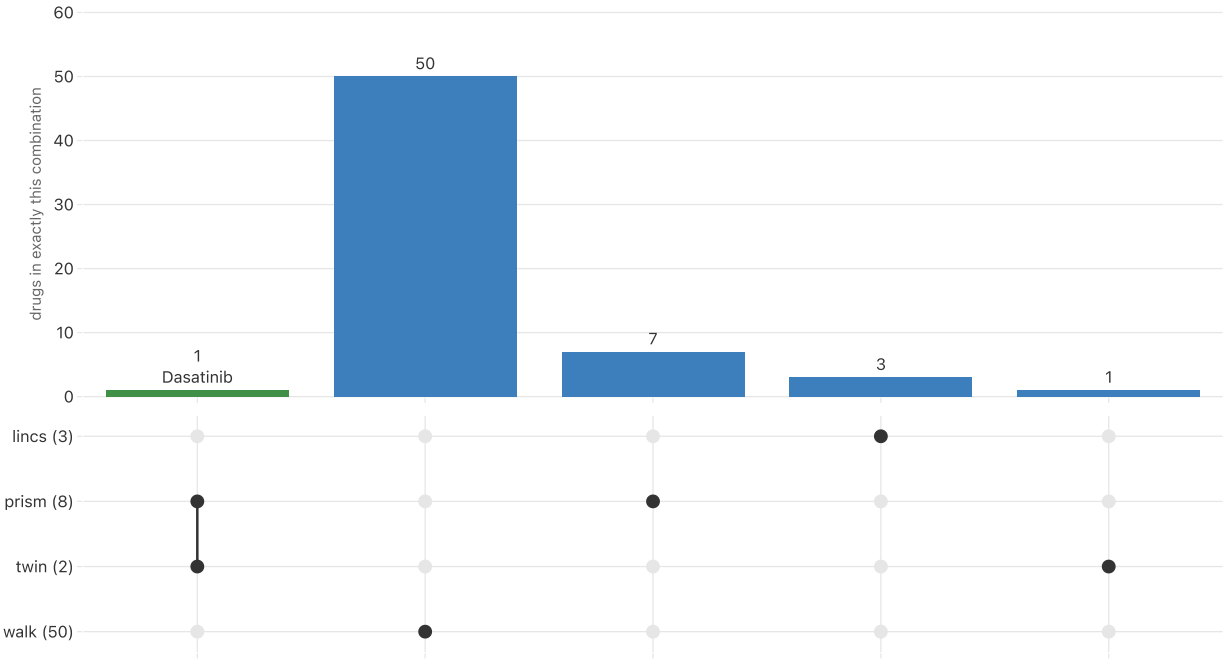
5.1 Do the generators agree more than chance?

A **permutation null** replaces each list with a random draw of the same size from its own universe, 10,000 times (seed 20260925), and counts how often ≥ 2 generators name the same drug.

test	observed	null mean	null 95th pct	p	role
all four generators	**1** (dasatinib)	0.30	1.0	**0.263**	the registered test (H094)
without dasatinib (registered, it was known before)	0	—	—	1	sensitivity
without G4 (failed control)	1	—	—	0.261	sensitivity

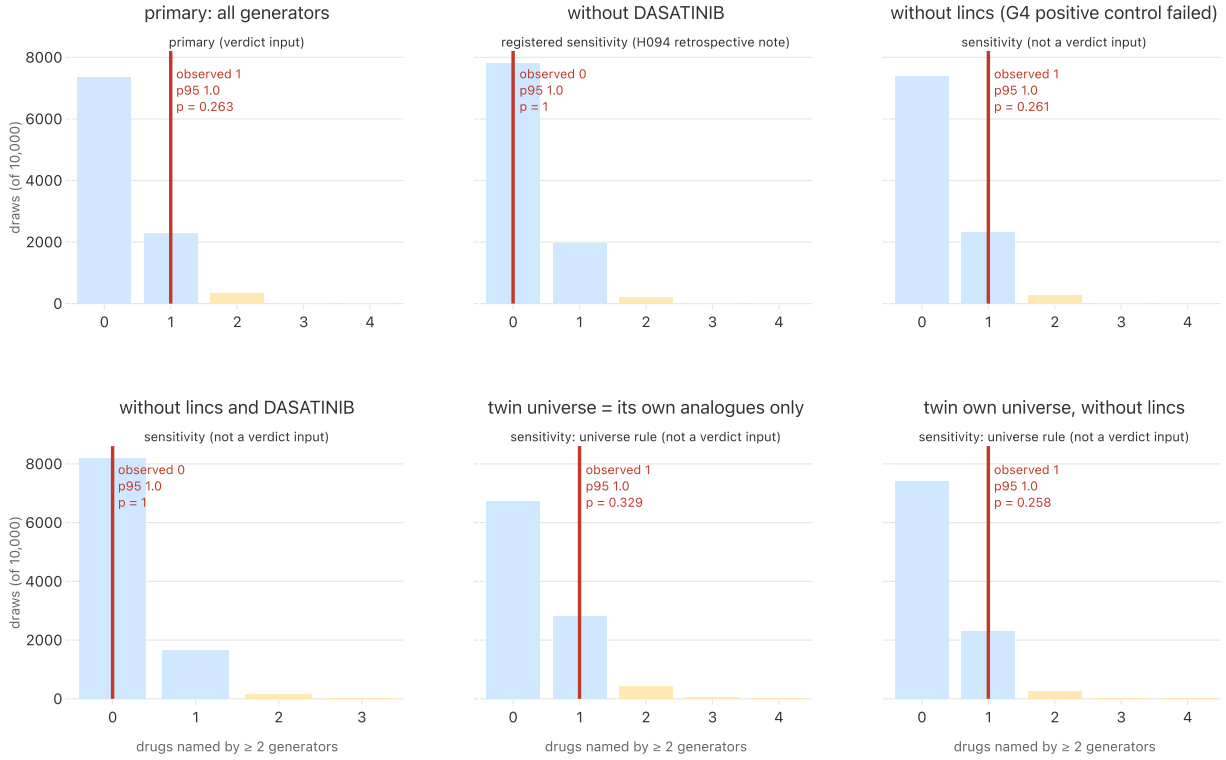
H094 is refuted (0.15): the four sources agree on one drug, which is what chance gives (entry 155).

Which generators name which approved drugs (62 drugs, k = 2)



> **Legend** green bar = a combination of ≥ 2 generators (a converging drug) · blue bar = one generator alone · dark dot = the generator is in the combination, grey = it is not · (n) = drugs the generator lists.

Convergence against a permutation null (each list redrawn from its own universe, seed 20260925)



> **Legend** blue bars = the null distribution · yellow = null counts above its 95th percentile (the bar the observed count had to clear) · solid line = observed: green if it clears the bar, red if not · only the first panel is a verdict input.

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6. The checks, and the candidates that survive them

> **Rubric:** Rigor and Impact. Every drug any generator named carries its check record.

check	rule	result over 62 drugs	source
K1 direction	a Mps1/TTK or Aurora inhibitor fails (A1); a tubulin drug carries a caution (A2)	32 pass · 30 caution (27 unannotated, 3 spindle poisons) · **0 fail**	entry 155
K2 the child's safety	fails only if **both** sources flag it: the child's genome (PharmCAT + CPIC guidelines) and the child's clinical terms against drug labels (OnSIDES)	51 pass (21 only because no label exists) · 10 caution · **1 fail: vincristine**	entry 155, method of entry 152
K3 contrary screen	fails if PRISM measures it pointing against its own direction	8 measured, all pass · 54 not applicable	entry 155

6.1 The survivor: dasatinib

	result
generators	twin #1 (D5: senescent-cell clearance delayed ageing in the BubR1 hypomorph, PMIDs 18516091, 22048312) and PRISM #8 (D2: aneuploidy contrast)
K1	pass: its mechanism (BCR-ABL, SRC, KIT, PDGFR, ephrin kinase inhibitor) is in no do-not class
K2	pass: *neither source flags it, which is not evidence of safety*. No pharmacogene guideline; 5 of 34 labels name rhabdomyolysis or weight loss (0.15, below the comparison set's 0.225)
K3	pass: aneuploidy Cliff's δ +0.135 (CI 0.024–0.243), one-sided p 0.0096
paediatric reports (FAERS, counts, no denominator)	growth retardation 15, malnutrition 2
H095	**supported, weakly (0.40)**: met as registered, but the convergence is chance-level, the drug was known before the join, and K3 re-reads its own generator's contrast

The checks: 1 converging drug(s), then each generator's top 10 (22 of 62 drugs; all in candidates.tsv)

	walk	twin	prism	lincs	K1 direction	K2 safety	K3 screen	survivor
Dasatinib		#1	#8		pass	pass	pass	yes
Phenytoin				#1	pass	caution	pass	no
Albendazole				#2	caution	caution	pass	no
Tofacitinib				#3	pass	pass	pass	no
Fomepizole			#1		pass	pass	pass	no
Cefpirome			#2		pass	pass	pass	no
Pizotifen			#3		pass	pass	pass	no
Fentiazac			#4		pass	pass	pass	no
Inositol Hexanicotinate			#5		caution	pass	pass	no
Delivert			#6		pass	pass	pass	no
Terazosin			#7		pass	pass	pass	no
Niacin		#2			pass	pass	pass	no
Fostamatinib	#1				pass	pass	pass	no
Theophylline	#2				pass	caution	pass	no
Nadh	#3				caution	pass	pass	no
Pyridoxal	#4				caution	pass	pass	no
Copper	#5				caution	pass	pass	no
Cannabidiol	#6				caution	caution	pass	no
L-Glutamine	#7				caution	pass	pass	no
Brentuximab Vedotin	#8				caution	caution	pass	no
Cyclosporine	#9				pass	pass	pass	no
Voclosporin	#10				pass	pass	pass	no

> **Legend** generator columns: blue = the generator names the drug (#rank) · K columns: green = pass, yellow = caution (kept, reason in the table), red = fail, grey = not run · survivor: green = converges and fails no check, red

> = does not.

6.2 The one drug both safety sources flag: vincristine

Vincristine is a standard rhabdomyosarcoma drug, so this is a monitoring question, not a removal. The genome half: an observed **CYP3A5 poor-metaboliser** call (CYP3A5 clears vincristine), read from 1 of its 7 positions in a region the callset barely covers. CEP72, the other vincristine-toxicity gene, is not covered by any caller. The phenotype half: vincristine's one label names **muscle atrophy**, an exact match to the child's own term, and paediatric FAERS holds 57 reports of vincristine with muscular weakness (entries 152, 155).

6.3 Every other candidate

The 61 other drugs are named by one generator each and carry no convergence. They are listed with their check record in Appendix A. The walk's list is dominated by highly connected nodes (vitamins, metals, amino acids): graph proximity without a mechanism argument. The screen's list sits at noise level ($q \geq 0.12$). The expression list is uninterpretable. None is argued here as a candidate.

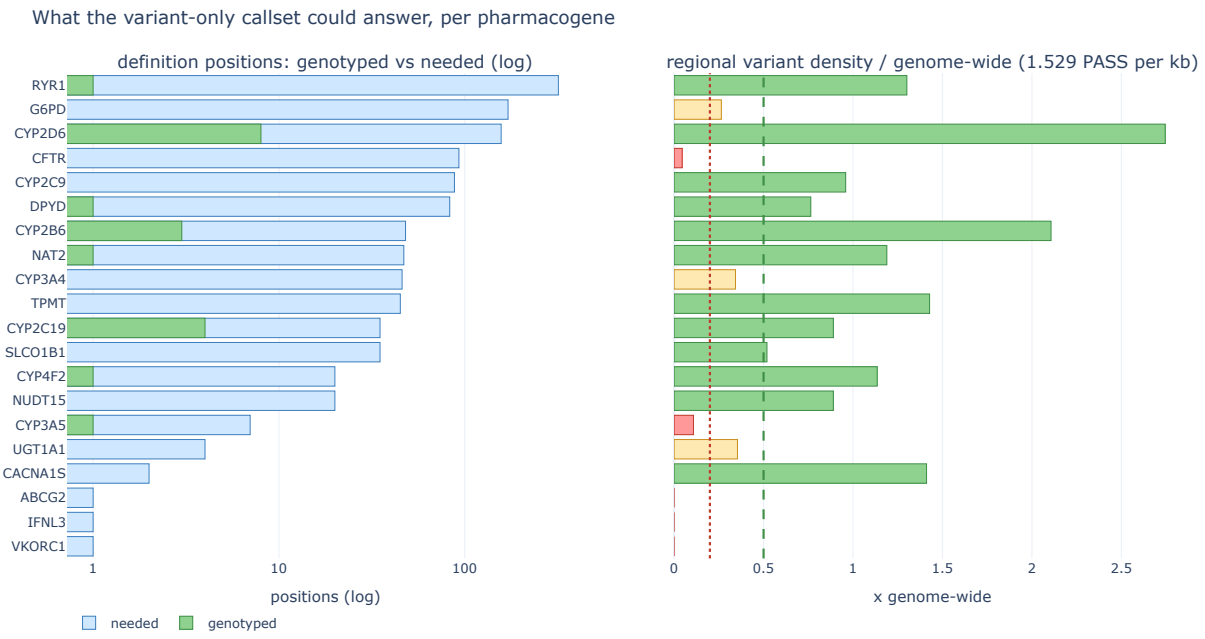
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7. What this could change: potential impact

> **Rubric:** Impact (25 %). What a treating team, other families and researchers could do next.

7.1 For this child: three questions for the treating team

1. **Order a targeted pharmacogenomic test.** The delivered callset lists variants only, so it answers **20 of 1,223** pharmacogene positions (**1.6 %**). Every gene with a prescribing guideline for a drug in this field is a **no call**: SLCO1B1 and ABCG2 (statins), CYP2C9, G6PD (chloroquine, hydroxychloroquine), UGT1A1 (irinotecan). A panel over those five, or a re-call of the sequencing reads at PharmCAT's positions, would answer it. The re-call needs the raw reads, not delivered as an alignment (≥ 1.7 h to stream 67.68 GB, entry 152).
2. **Vincristine**: the two-source flag in §6.2.
3. **Anaesthesia**: an incidental RYR1 call of *uncertain* malignant-hyperthermia susceptibility, in a child likely to face repeated anaesthesia (entry 152).



> **Legend** left: blue = positions PharmCAT needs · green = positions this callset genotyped · right: green = region called at ≥ 0.5× the genome-wide rate · yellow = 0.2–0.5× (a decision point) · red = below 0.2×, where assuming "no variant = reference" is not supported.

7.2 For preclinical testing: which mouse to use

Any candidate here should be tested in the BubR1 **hypomorph** (H/H or H/L1002P), the child's phenotype twin, not in a knockout or heterozygous null (entry 150). No approved drug has been tested in that mouse. The twin has no kidney phenotype recorded, so the child's nephrocalcinosis is unexplained by the model and is itself a readout worth measuring.

7.3 The experiment that settles the mechanism class

A BubR1 Western blot with a cycloheximide chase (the protein's decay rate) in the child's cells against the parents' would settle "unstable or inactive?" directly and override every prediction in §3.3. A positive answer (unstable) is also the biomarker for direction D3 (entry 149).

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8. What is new here, against standard practice

> **Rubric:** Innovation (25 %).

idea	standard practice	what iteration 4 did	source
convergence **tested**	"several sources agree" is asserted	independent generators, rules fixed before the join, a permutation null; the refutation is reported	entry 155
generators vs checks	one pipeline both proposes and filters	a check can never vote for a drug it could have generated	step 10 README
patient → mouse **genotype**	disease → gene lists	the child's terms against all 76,699 MGI genotypes, with an information-matched null; the twin's **allele class** decides which rescues transfer	entry 150
calibrate before placing	score the patient's variant	an all-substitution stability atlas, tested on published allele classes first; the negative calibration is the finding	entry 149
per-child, two-source safety	label warnings for children in general	the child's **genome** and **clinical terms** side by side per drug; "unflagged" worded so it can never read as "safe"; a regional callability measure says where "no call = reference" is defensible	entries 152, 155
pre-registration and an append-only register	results written after the fact	the 12 claims in §12 filed before their runs; every verdict a new revision, never an edit	`docs/science-register.json`

9. Scalability: the same chain, a second disease

> **Rubric:** Scalability (15 %).

Every stage is a tool with one table contract and a provenance manifest. The inputs are meant to be a gene, a disease node, a list of clinical terms and, for the safety check, a VCF. Step 11 measured how far that holds.

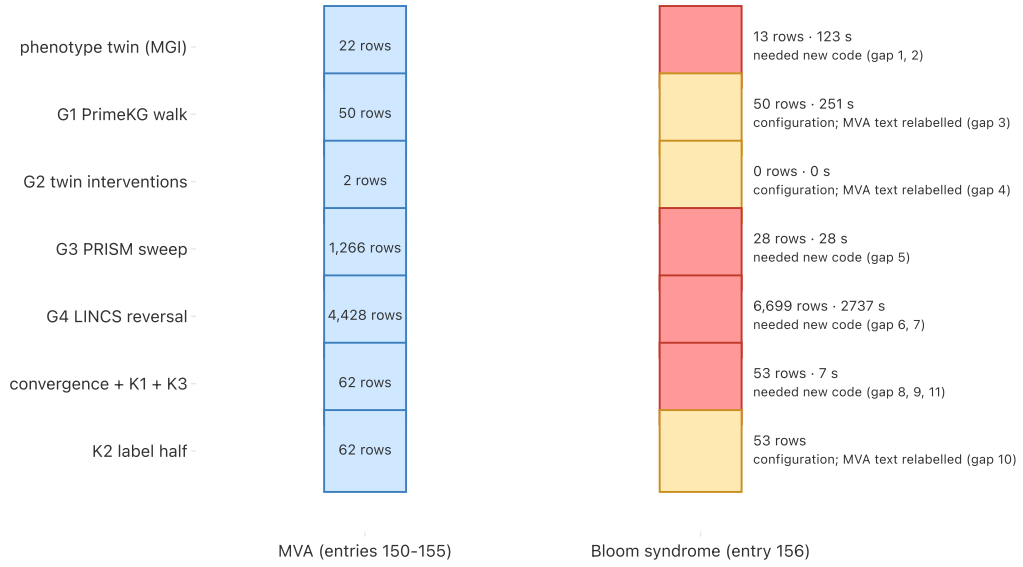
stage	tool	what changes for another case
phenotype twin	`tools/phenotype_twin.py`	the HPO terms and the gene
PrimeKG walk	`mva.drugs.graph_walk`	the disease node
screen sweep	`tools/screen_sweep.py`	the lineage and covariate contrasts
signature reversal	`tools/signature_reversal.py`	the gene and its positive-control siblings
convergence + checks	`tools/converge.py`	generator ids and paths
per-child safety	`tools/patient_safety.py`	the VCF, the drug table, the HPO terms

The second case (step 11, Bloom syndrome, `BLM`). Bloom syndrome swaps the gene (a DNA helicase), the pathway (DNA repair, not the mitotic checkpoint) and the clinical terms (42, from the public `phenotype.hpoa`), while asking the same kind of question (entry 156).

stage	on Bloom syndrome	rows written	what it took
phenotype twin	best *Blm* mouse at 1.15 % of 76,699 (retrospective: entry 150 ran this control; *Recql4* ranks higher)	13 genotypes	new code (its output directory was fixed)
G1 PrimeKG walk	1,026 drugs reached, 239 labelled; KIT dominates Bloom's disease node	50	flags, but MVA text in its manifest
G2 twin rescues	no *Blm*-mouse rescue names an approved drug	0	flags, from a curated table
G3 PRISM sweep	only 14 Launched drugs are measured in blood-cancer lines	28	new code (lineage in constants)
G4 LINCS reversal	positive control (RMI1, RMI2, TOP3A) failed, as for MVA	6,699	new code (gene in constants)
convergence + checks	no drug named by ≥ 2 generators (p 1.0); K1 fails cisplatin	53	new code (K1 rule in constants)

H096 is refuted (0.15). The chain runs on a second disease and proposes a different set (Jaccard 0.095 against a 0.20 bar), but not by configuration alone. The driver `tools/transfer_it4.py` (378 lines, no existing line edited) fills the four code gaps and corrects 136 strings of first-case text by recorded substitutions, re-hashing every manifest. That list is the work order for making the tools truly case-agnostic.

The chain on a second case: which stages ran by configuration alone



> **Legend** blue = the MVA run, built for that case · Bloom column: green = configuration alone, yellow = ran from flags but wrote MVA-case text that `relabel` had to correct, red = needed new code, grey = not written · "gap" = entry 156's static-survey row · seconds = wall time.

9.1 What it cost

stage	compute	cost	source
GPU variant models (NT-v2, Evo 2, ESM-C, Boltz-2)	Modal	$14.57	entry 148
stability atlas (ThermoMPNN)	Modal T4, 16 GPU-s	$0.015	entry 149
phenotype twin	laptop, 53 s	$0	entry 150
PRISM replication and sweep	laptop, ~4 min	$0	entries 151, 154
per-child safety	laptop, < 1 min PharmCAT	$0	entry 152
LINCS reversal	public APIs	$0	entry 153
the second disease, all stages	laptop + public APIs, 53 min	$0	entry 156
convergence + checks	laptop, 27 s	$0	entry 155

---

10. What this report cannot tell you

> **Rubric:** Rigor. The limits, next to the results they bound.

- **The drug list is thin.** Four generators agree on one drug, at chance level (H094 refuted). Dasatinib is a hypothesis with two independent routes, not a finding.
- **Two generators are weak.** The screen sweep's rhabdomyosarcoma arm rests on three cell lines for 619 of 633 compounds; its "approved" label covers only compounds the 2020 release classified (entry 154). The expression generator failed its positive control (entry 153).
- **The walk rewards connectivity.** Vitamins and metals rank high because they touch many proteins (entry 142).
- **The mechanism class is predicted, not measured.** ΔΔG cannot separate published allele classes at n = 8 (entry 149); the Boltz-2 fold is inconclusive (entry 147).
- **The twin's rescues are genetic**, not pharmacological; no approved drug was tested in a BubR1-deficient mouse (entry 150).
- **Safety sources are silent more often than they speak.** 22 of 62 candidates have no OnSIDES label; the genome answers 1.6 % of the positions (entries 152, 155).
- **The chain is not yet configuration-only.** A second disease needed a 378-line driver (entry 156).
- **Nothing here was tested in this child.**

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11. Reproducibility

> **Rubric:** Rigor and Scalability.

Code: `https://github.com/agharsallah/mva-hackathon` (public; no patient data, no genomic position in any tracked file). Every table under `ressources/track2/it4/` carries a `.provenance.json` (inputs with URL, sha256 and licence; parameters; the entry that produced it). The commands are in each entry's `## Commands` section. In order:

```
uv run python tools/phenotype_twin.py run --gene Bub1b          # entry 150
uv run python tools/screen_sweep.py run                         # entry 154
uv run python tools/signature_reversal.py run                   # entry 153
uv run python tools/converge.py twin ...                        # entry 155 (full flags there)
uv run python tools/converge.py run ...                         # entry 155
uv run python tools/render_step10_s2.py
```

12. Where every claim lives

claim	test	verdict	entry
H074	the walk's top 20 differs from a decoy gene's walk (overlap ≤ 5 of 20)	supported 0.75 (overlap 2 of 20)	142
H083	stability separates published allele classes	refuted 0.15	149
H084	the child's missense is in the top 20 % of ΔΔG	supported 0.55	149
H085	the child's terms pick a Bub1b mouse in the top 5 %	supported 0.70	150
H086	the best Bub1b mouse is a hypomorph	testing 0.45	150
H089	the D5 drugs, reported whichever way they point	supported 0.80	151
H090	the child's genome carries an actionable pharmacogenomic result	supported 0.40	152
H092	a BUB1B-loss signature is specifically reversed by an approved drug	testing 0.50 (uninterpretable)	153
H093	a Launched drug is selective for both aneuploid and RMS lines	refuted 0.10	154
H094	the four generators converge beyond chance	refuted 0.15	155
H095	a converging drug survives every check	supported 0.40	155
H096	the chain runs on a second disease by configuration	refuted 0.15 (it runs, with a 378-line driver)	156

Entries are under `docs/iteration4/step-<n>/methodology/`. Three iteration-4 claims are left out on purpose: H087, H088 and H091 tested drugs drawn from an earlier iteration's shortlist. Their *methods* are reused here (entries 151, 152); their *candidate rows* are not. Entry 151's screen-wide Mps1/Aurora measurement (§4, A1) is independent of any list and is kept.

Appendix A — every drug any generator named

From entry 155 §5 (`ressources/track2/it4/candidates.tsv`). "no Hub MOA": the Drug Repurposing Hub does not annotate the drug's mechanism. "no OnSIDES label": unflagged only because no label exists. K3 `n/a`: no direction maps to a PRISM contrast.

drug	generators (rank)	direction hint	K1 direction	K2 safety	K3 screen	survivor
Dasatinib	twin #1, prism #8	D2, D5	pass	pass	pass (aneuploidy δ +0.135)	**yes**
Acyclovir	walk #45	none	pass	pass	n/a	no
Adenosine	walk #26	none	pass	pass	n/a	no
Albendazole	lincs #2	none	caution: spindle poison	caution (phenotype)	n/a	no
Aldesleukin	walk #36	none	caution: no Hub MOA	pass	n/a	no
Basiliximab	walk #37	none	caution: no Hub MOA	pass	n/a	no
Belinostat	walk #15	none	pass	caution (genome)	n/a	no
Bleomycin	walk #40	none	caution: no Hub MOA	pass	n/a	no
Brentuximab Vedotin	walk #8	none	caution: no Hub MOA	caution (phenotype)	n/a	no
Cabazitaxel	walk #47	none	caution: spindle poison	pass (no OnSIDES label)	n/a	no
Caffeine	walk #49	none	pass	pass	n/a	no
Calcium	walk #39	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Cannabidiol	walk #6	none	caution: no Hub MOA	caution (phenotype)	n/a	no
Carbamazepine	walk #29	none	pass	caution (genome)	n/a	no
Cefpirome	prism #2	D2	pass	pass (no OnSIDES label)	pass (aneuploidy δ +0.188)	no
Citrulline	walk #43	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Cladribine	walk #32	none	pass	pass	n/a	no
Copper	walk #5	none	caution: no Hub MOA	pass	n/a	no
Cyclosporine	walk #9	none	pass	pass	n/a	no
Delivert	prism #6	D2	pass	pass (no OnSIDES label)	pass (aneuploidy δ +0.142)	no
Denileukin Diftitox	walk #38	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Doxorubicin	walk #30	none	pass	pass	n/a	no
Dronedarone	walk #22	none	pass	pass	n/a	no
Fentiazac	prism #4	D2	pass	pass (no OnSIDES label)	pass (aneuploidy δ +0.143)	no
Fomepizole	prism #1	D4	pass	pass (no OnSIDES label)	pass (rms δ +0.925)	no

drug	generators (rank)	direction hint	K1 direction	K2 safety	K3 screen	survivor
Fostamatinib	walk #1	none	pass	pass	n/a	no
Glycine	walk #19	none	caution: no Hub MOA	pass	n/a	no
Inositol Hexanicotinate	prism #5	D4	caution: no Hub MOA	pass (no OnSIDES label)	pass (rms δ +0.877)	no
Isoleucine	walk #27	none	caution: no Hub MOA	pass	n/a	no
L-Glutamine	walk #7	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Lenalidomide	walk #25	none	pass	caution (phenotype)	n/a	no
Levothyroxine	walk #16	none	pass	pass	n/a	no
Liothyronine	walk #17	none	pass	pass	n/a	no
Magnesium	walk #41	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Magnesium Carbonate	walk #42	none	caution: no Hub MOA	pass	n/a	no
Mecasermin	walk #23	none	caution: no Hub MOA	pass	n/a	no
Micafungin	walk #21	none	pass	pass	n/a	no
NADH	walk #3	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Niacin	twin #2	D3	pass	pass	n/a	no
Perindopril	walk #33	none	pass	pass	n/a	no
Phenytoin	lincs #1	none	pass	caution (genome)	n/a	no
Pizotifen	prism #3	D4	pass	pass (no OnSIDES label)	pass (rms δ +0.915)	no
Podofilox	walk #48	none	caution: no Hub MOA	pass	n/a	no
Pyridoxal	walk #4	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Pyruvic Acid	walk #46	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Quercetin	walk #20	none	pass	pass (no OnSIDES label)	n/a	no
Romidepsin	walk #34	none	pass	pass	n/a	no
Sacituzumab Govitecan	walk #50	none	caution: no Hub MOA	caution (genome)	n/a	no
Serine	walk #28	none	caution: no Hub MOA	pass	n/a	no
Sorafenib	walk #31	none	pass	caution (phenotype)	n/a	no

drug	generators (rank)	direction hint	K1 direction	K2 safety	K3 screen	survivor
Taurine	walk #24	none	pass	pass	n/a	no
Tenecteplase	walk #44	none	caution: no Hub MOA	pass	n/a	no
Terazosin	prism #7	D4	pass	pass	pass (rms δ +0.871)	no
Theophylline	walk #2	none	pass	caution (phenotype)	n/a	no
Thyroid, Porcine	walk #18	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Tofacitinib	lincs #3	none	pass	pass (no OnSIDES label)	n/a	no
Vincristine	walk #13	none	caution: spindle poison	**fail** (both sources)	n/a	no
Vitamin A	walk #14	none	caution: no Hub MOA	pass	n/a	no
Voclosporin	walk #10	none	pass	pass	n/a	no
Vorinostat	walk #35	none	pass	pass (no OnSIDES label)	n/a	no
Zinc	walk #11	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no
Zinc Sulfate, Unspecified Form	walk #12	none	caution: no Hub MOA	pass (no OnSIDES label)	n/a	no